Extra Nucleus

It is the outer part of an atom which contains electrons only. An electron is a negatively charged particle having negligible mass. The charge of an electron is equal but opposite to that of a proton. Also the number of electrons are equal to the number of protons in an atom under ordinary conditions. Therefore an atom is neutral as a 'whole'. The number of electrons or protons in an atom is called atomic number.

3.1.2 Atomic Number

Normally at ordinary conditions the number of electrons in the extra nucleus (i.e, orbits) are equal to number of protons present in the nucleus.

Therefore, an atom is neutral as a whole. The number of electrons or protons in an atom is called atomic number.

$\therefore$ *Atomic Number = No. of Protons (or) No. of Electrons in an atom.*

3.1.3 Atomic Weight

The sum of the protons and neutrons decides the entire weight of an atom and is called atomic weight. The electrons are not taken for consideration because it is having negligible mass as compared to protons or neutrons.

Atomic Weight = No. of Protons + No. of Neutrons

The electrons present in the atom is root cause for the action of any type of conduction (say electricity, heat etc).The electrons in an atom revolve around the nucleus in different orbit or paths. The number and arrangement of electrons in any orbit is determined by the following rules.

1. The number of electron in any orbit is given by $2n^2$ where n is the number of orbit.

For Example:

First orbit contains $2x1^2 = 2$ electrons
Second orbit contains $2x2^2 = 8$ electrons
Third orbit contains $2x3^2 = 18$ electrons and so on

2. The last orbit cannot have more than 8 electrons.
3. The last but one orbit cannot have more than 18 electrons.

3.2. STRUCTURE OF ELEMENTS

We have seen all atoms are made up of protons, neutrons and electrons. The difference between types of elements is due to the different number and arrangement of these particles within their atoms. The structure of copper atom is different from carbon atom and hence the two elements have different properties.

The atomic structure can be easily drawn if we know the atomic weight and atomic number of an element.

For example:

We take copper atom,

Atomic weight = 64
Atomic number = 29
No. of Protons = No of Electrons = 29 and
No of Neutrons = 64-29=35.

ELEC_EQUIP_Chapter03.indd 62 11/10/2020 11:36:39 AM